

Costruzione e training di una rete neurale MLP

Obiettivo è imparare a stimare il prezzo di una casa conoscendo:

- la superficie in metri quadrati;
- il numero di stanze.

Creiamo un piccolo dataset "giocattolo" con 8 case.

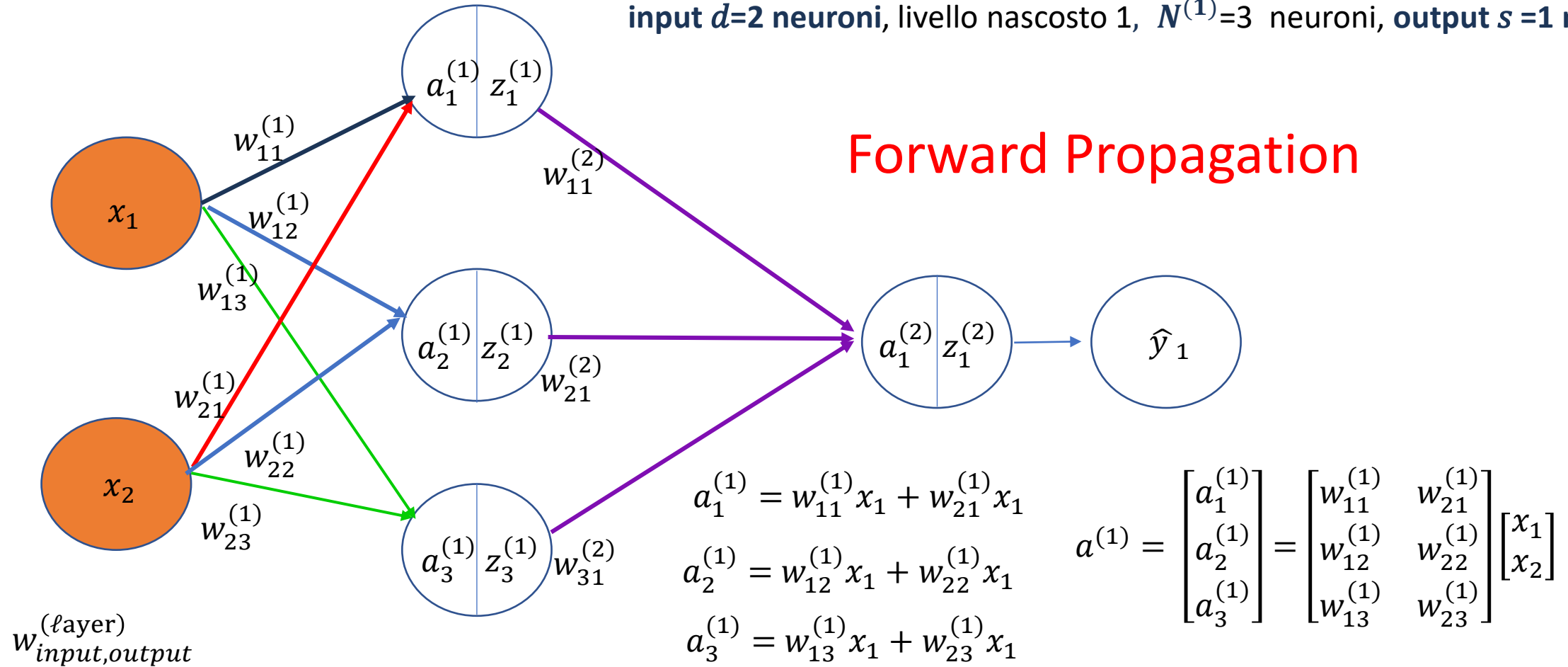
Ogni riga di X descrive una casa:

- colonna 0 $\rightarrow$ superficie in m^2 ;
- colonna 1 $\rightarrow$ numero di stanze.

Il vettore y contiene il prezzo reale di ogni casa (in migliaia di €). Questi sono i valori che la rete deve imparare a prevedere.

rete a **3** livelli $d:N^{(1)}:s$

input $d=2$ neuroni, livello nascosto 1, $N^{(1)}=3$ neuroni, output $s=1$ neuroni



$$z_i^{(0)} = x_i, \quad i = 1, \dots, d$$

$$\hat{y}_i = z_i^{(L+1)}, \quad i = 1, \dots, s$$

$$a_i^{(\ell)} = \sum_{j=1}^{N^{(\ell-1)}} w_{ji}^{(\ell)} z_j^{(\ell-1)}$$

$$z_i^{(\ell)} = f(a_i^{(\ell)})$$

f : activation function: funzione **derivabile non lineare**.

$$W^{(1)} = \begin{bmatrix} w_{11}^{(1)} & w_{21}^{(1)} \\ w_{12}^{(1)} & w_{22}^{(1)} \\ w_{13}^{(1)} & w_{23}^{(1)} \end{bmatrix} \in R^{N^{(1)} \times d}$$

$$a^{(1)} = W^{(1)}z^{(0)}$$

$$z^{(1)} = f'(a^{(1)})$$

$$W^{(2)} = \begin{bmatrix} w_{11}^{(2)} & w_{21}^{(2)} & w_{31}^{(2)} \end{bmatrix} \in R^{s \times N^{(1)}}$$

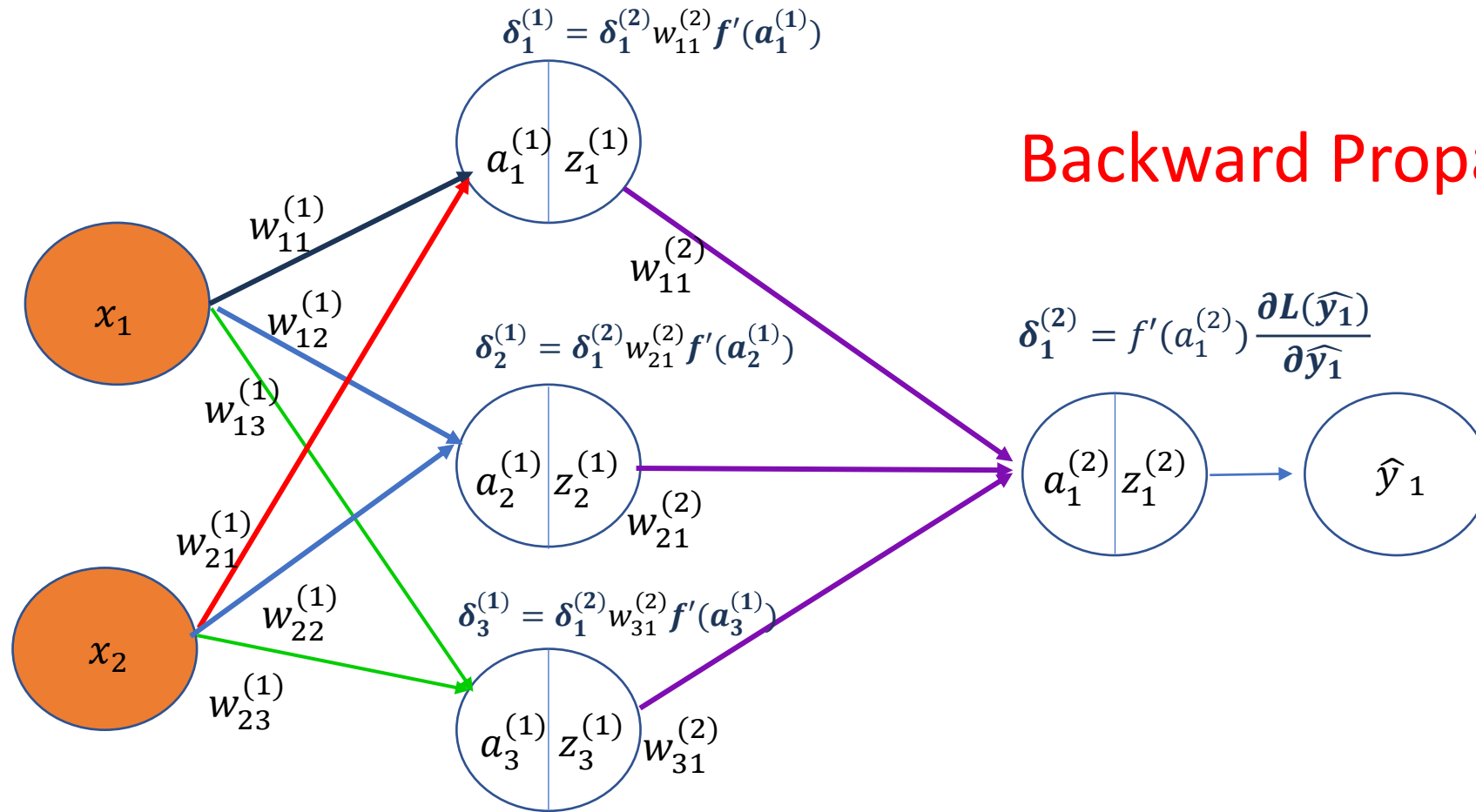
$$a^{(2)} = \begin{bmatrix} a_1^{(2)} \end{bmatrix} = \begin{bmatrix} w_{11}^{(2)} & w_{21}^{(2)} & w_{31}^{(2)} \end{bmatrix} \begin{bmatrix} z_1^{(1)} \\ z_2^{(1)} \\ z_3^{(1)} \end{bmatrix}$$

$$z_1^{(2)} = f'(a^{(2)})$$

$$z_i^{(0)} = x_i, \quad i = 1, \dots, d$$

$$a^{(1)} = W^{(1)}z^{(0)}$$

Backward Propagation



$$\frac{\partial L(w)}{\partial w_{11}^{(2)}} = \delta_1^{(2)} z_1^{(1)}$$

$$\frac{\partial L(w)}{\partial w_{21}^{(2)}} = \delta_1^{(2)} z_2^{(1)}$$

$$\frac{\partial L(w)}{\partial w_{31}^{(2)}} = \delta_1^{(2)} z_3^{(1)}$$

Propagazione del delta al layer nascosto

Il simbolo $\odot$ indica il **prodotto elemento per elemento**

$$\begin{bmatrix} \delta_1^{(1)} \\ \delta_2^{(1)} \\ \delta_3^{(1)} \end{bmatrix} = \delta_1^{(2)} \begin{bmatrix} w_{11}^{(2)} \\ w_{21}^{(2)} \\ w_3^{(2)} \end{bmatrix} \odot \begin{bmatrix} f'(a_1^{(1)}) \\ f'(a_2^{(1)}) \\ f'(a_3^{(1)}) \end{bmatrix}$$

Derivate parziali rispetto ai pesi che collegano il layer nascosto al layer di input

$$\frac{\partial L(w)}{\partial w_{11}^{(1)}} = \delta_1^{(1)} z_1^{(0)}$$

$$\frac{\partial L(w)}{\partial w_{21}^{(1)}} = \delta_1^{(1)} z_2^{(0)}$$

$$\frac{\partial L(w)}{\partial w_{12}^{(1)}} = \delta_2^{(1)} z_1^{(0)}$$

$$\frac{\partial L(w)}{\partial w_{22}^{(1)}} = \delta_2^{(1)} z_2^{(0)}$$

$$\frac{\partial L(w)}{\partial w_{13}^{(1)}} = \delta_3^{(1)} z_1^{(0)}$$

$$\frac{\partial L(w)}{\partial w_{23}^{(1)}} = \delta_3^{(1)} z_2^{(0)}$$

Forward propagation per n campioni – estensione matriciale

$$A_1 \in \mathbb{R}^{3 \times n} = \begin{bmatrix} w_{11}^{(1)} & w_{21}^{(1)} \\ w_{12}^{(1)} & w_{22}^{(1)} \\ w_{13}^{(1)} & w_{23}^{(1)} \end{bmatrix} \begin{bmatrix} z_{11}^{(0)} & z_{21}^{(0)} & \dots & z_{n1}^{(0)} \\ z_{21}^{(0)} & z_{22}^{(0)} & \dots & z_{n,1}^{(0)} \end{bmatrix}$$

$$Z_1 = f'(A_1)$$

$$A_2 \in \mathbb{R}^{1 \times n} = \begin{bmatrix} w_{11}^{(2)} & w_{21}^{(2)} & w_{31}^{(2)} \end{bmatrix} \begin{bmatrix} z_{11}^{(1)} & z_{21}^{(1)} & \dots & z_{n1}^{(1)} \\ z_{21}^{(1)} & z_{22}^{(1)} & \dots & z_{n,1}^{(1)} \end{bmatrix}$$

$$Z_2 = f'(A_2)$$

Backpropagation per n campioni — estensione matriciale

Da 1 campione a n campioni: la funzione costo diventa una media

Funzione costo (MSE)

$$C(w) = (1/n) \cdot \sum_r L^{(r)} \quad \text{dove} \quad L^{(r)} = \frac{1}{2} (\hat{y}^{(r)} - y^{(r)})^2$$

$\delta_1^{(2)}$ — delta del neurone di output (shape: $1 \times n$)

1 campione (dalle slide):

$$\delta_1^{(2)} = f'(a_1^{(2)}) \times \partial L / \partial \hat{y}$$

$$\text{dove } \partial L / \partial \hat{y} = 2(\hat{y} - y)$$

general
izza →

n campioni (forma matriciale):

$$\delta_1^{(2)} = \frac{1}{n} (\hat{Y} - Y) \odot f'(A_2)$$

$\odot$ = prodotto elemento per elemento (Hadamard)

Gradienti di W_2 e b_2

1 campione:

$$\partial L / \partial w_{11}^{(2)} = \delta_1^{(2)} z_1^{(1)}$$

$$\partial L / \partial w_{21}^{(2)} = \delta_1^{(2)} z_2^{(1)}$$

$$\partial L / \partial w_{31}^{(2)} = \delta_1^{(2)} z_3^{(1)}$$

n campioni:

$$\partial C / \partial W_2 = \delta_1^{(2)} @ Z_1^T \quad \text{shape: } (1,n) @ (n,3) = (1,3)$$

$$\partial C / \partial b_2 = \sum_r \delta_1^{(2,r)} = \text{delta}_1^{(2)}.sum(axis=1) \quad \text{shape: } (1,1)$$

Backpropagation per n campioni — layer nascosto

Propagazione del delta al layer nascosto e gradienti di W_1 , b_1

$\delta_i^{(1)}$ — delta dei neuroni hidden (shape: $3 \times n$)

1 campione (dalle slide):

$$\delta_1^{(1)} = \delta_1^{(2)} w_{11}^{(2)} f'(a_1^{(1)})$$

$$\delta_2^{(1)} = \delta_1^{(2)} w_{21}^{(2)} f'(a_2^{(1)})$$

$$\delta_3^{(1)} = \delta_1^{(2)} w_{31}^{(2)} f'(a_3^{(1)})$$

Forma vettoriale (slide 4):

$$[\delta_1 \delta_2 \delta_3]^{(1)} = \delta_1^{(2)} [w_{11} w_{21} w_{31}]^{(2)} \odot [f'(a_1) f'(a_2) f'(a_3)]^{(1)}$$

n campioni (forma matriciale):

$$\delta^{(1)} = W_2^T @ \delta_1^{(2)} \odot f'(A_1)$$

il prodotto matriciale somma automaticamente i contributi degli nn n campioni, sostituendo il loop esplicito

shape: $(3,1) @ (1,n) \rightarrow (3,n)$ poi $\odot (3,n)$

Perché W_2^T ? Nella forward pass W_2 comprimeva da 3 (hidden) a 1 (output). Nella backward W_2^T espande da 1 a 3, distribuendo l'errore a ciascun neurone hidden.

$\odot f'(A_1)$: dove il neurone era spento ($a_i^{(1)} \leq 0$), $f' = 0 \rightarrow$ il gradiente viene bloccato.

Gradienti di W_1 e b_1

1 campione (dalle slide):

$$\partial C / \partial w_{11}^{(1)} = \delta_1^{(1)} z_1^{(0)} \quad \partial C / \partial w_{21}^{(1)} = \delta_1^{(1)} z_2^{(0)}$$

$$\partial C / \partial w_{12}^{(1)} = \delta_2^{(1)} z_1^{(0)} \quad \partial C / \partial w_{22}^{(1)} = \delta_2^{(1)} z_2^{(0)}$$

$$\partial C / \partial w_{13}^{(1)} = \delta_3^{(1)} z_1^{(0)} \quad \partial C / \partial w_{23}^{(1)} = \delta_3^{(1)} z_2^{(0)}$$

n campioni:

$$\partial C / \partial W_1 = \delta^{(1)} @ Z_0^T \quad \text{shape: } (3,n) @ (n,2) = (3,2)$$

$$\partial C / \partial b_1 = \delta^{(1)}.sum(axis=1) \quad \text{shape: } (3,1)$$